

Module 2 Activity 3: Working with Functions

Purpose:

Create a list of datasets inside a geodatabase, filter the list using wildcards and run a geoprocessing tool based on the results of the generated list.

Instructions:

Getting a list of feature classes in a workspace using the **ListFeatureClasses()** function

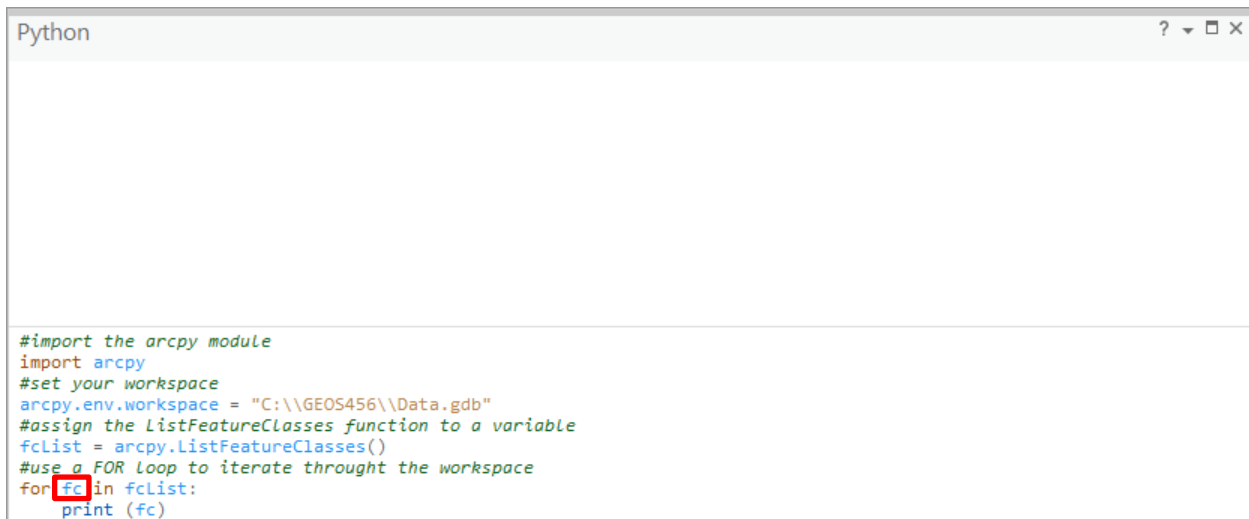
1. Import ArcPy
2. Set the workspace

NOTE: You must always set the workspace before calling any list function. If you do not tell the script where to look, it will not be able to find any data.

3. Create a variable called **fcList**
4. Call the **ListFeatureClasses()** function and assign it to the variable

Leaving the brackets empty in the function will generate a complete list. Specifying parameters inside the brackets will limit or restrict what is generated.

5. Use a **for** loop to find all the feature classes in the **fcList**



```
#import the arcpy module
import arcpy
#set your workspace
arcpy.env.workspace = "C:\\\\GEOS456\\\\Data.gdb"
#assign the ListFeatureClasses function to a variable
fcList = arcpy.ListFeatureClasses()
#use a FOR loop to iterate through the workspace
for fc in fcList:
    print (fc)
```

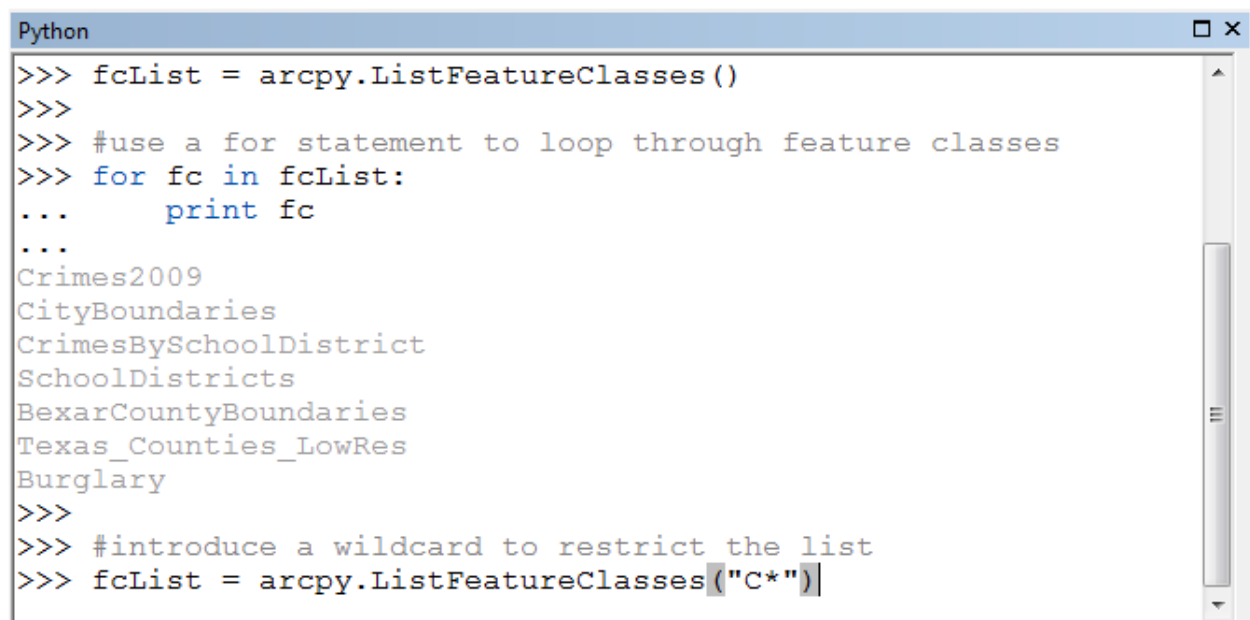
In the above screenshot, **fc** (highlighted in red) is the name of a variable specified in the **for loop**. For each iteration of the loop, this variable is assigned a different value.

6. Run the script by pressing **enter** twice
7. Note the result in the Python Window



```
Python
#import the arcpy module
import arcpy
#set your workspace
arcpy.env.workspace = "C:\\\\GEOS456\\\\Data.gdb"
#assign the ListFeatureClasses function to a variable
fcList = arcpy.ListFeatureClasses()
#use a FOR loop to iterate through the workspace
for fc in fcList:
    print (fc)
Crimes2009
CityBoundaries
CrimesBySchoolDistrict
SchoolDistricts
BexarCountyBoundaries
Texas_Counties_LowRes
Burglary
```

8. Restrict which values are returned by adding a wildcard.

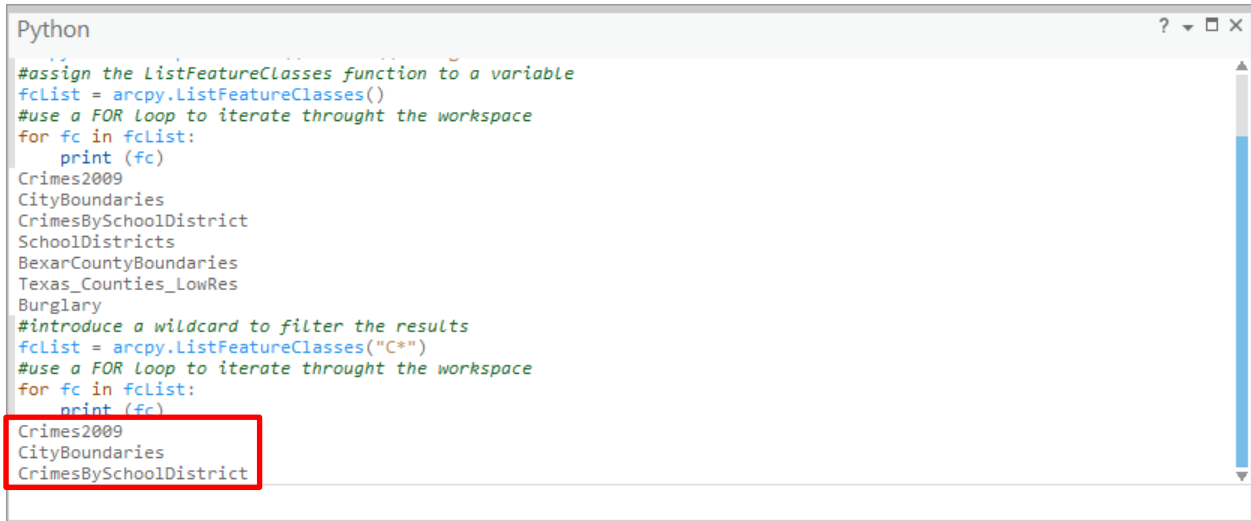


```
Python
>>> fcList = arcpy.ListFeatureClasses()
>>>
>>> #use a for statement to loop through feature classes
>>> for fc in fcList:
...     print fc
...
Crimes2009
CityBoundaries
CrimesBySchoolDistrict
SchoolDistricts
BexarCountyBoundaries
Texas_Counties_LowRes
Burglary
>>>
>>> #introduce a wildcard to restrict the list
>>> fcList = arcpy.ListFeatureClasses("C*")
```

A wildcard character (*****) is introduced to the brackets of the function to restrict the values returned. In the above screenshot, only feature classes that start with the letter **C** will be listed.

9. Use the **for** loop again to find all the datasets that start with **C**

10. Press **enter** twice to generate the resulting list



```
Python
#assign the ListFeatureClasses function to a variable
fclist = arcpy.ListFeatureClasses()
#use a FOR loop to iterate through the workspace
for fc in fclist:
    print (fc)
Crimes2009
CityBoundaries
CrimesBySchoolDistrict
SchoolDistricts
BexarCountyBoundaries
Texas_Counties_LowRes
Burglary
#introduce a wildcard to filter the results
fclist = arcpy.ListFeatureClasses("C*")
#use a FOR loop to iterate through the workspace
for fc in fclist:
    print (fc)
Crimes2009
CityBoundaries
CrimesBySchoolDistrict
```

The same method can be used to restrict the values by feature type. The feature type can be added to the brackets of the function to return either **Point**, **Polyline**, **Polygon** features.

How would you restrict the list to return only feature classes that start with the letter **C** and are **Polygon** feature type?

Once a list is generated, you can continue with geoprocessing by using the result of the **ListFeatureClasses()** function as the inputs of a geoprocessing tool.

11. Use the **buffer** tool to buffer one of the results from the list

The buffer tool generates the desired output in the workspace and is added to Pro

```
Python
CityBoundaries
CrimesBySchoolDistrict
SchoolDistricts
BexarCountyBoundaries
Texas_Counties_LowRes
Burglary
#introduce a wildcard to filter the results
fcList = arcpy.ListFeatureClasses("C*")
#use a FOR loop to iterate through the workspace
for fc in fcList:
    print (fc)
Crimes2009
CityBoundaries
CrimesBySchoolDistrict

#buffer the boundaries by 100m
arcpy.analysis.Buffer("CityBoundaries", "CityBound_Buffer100m", "100 Meters")
<Result 'C:\GEOS456\Data.gdb\CityBound_Buffer100m'>
```

